Technical appendix: enrollment uncertainty and the proposed closure of Mesa Elementary

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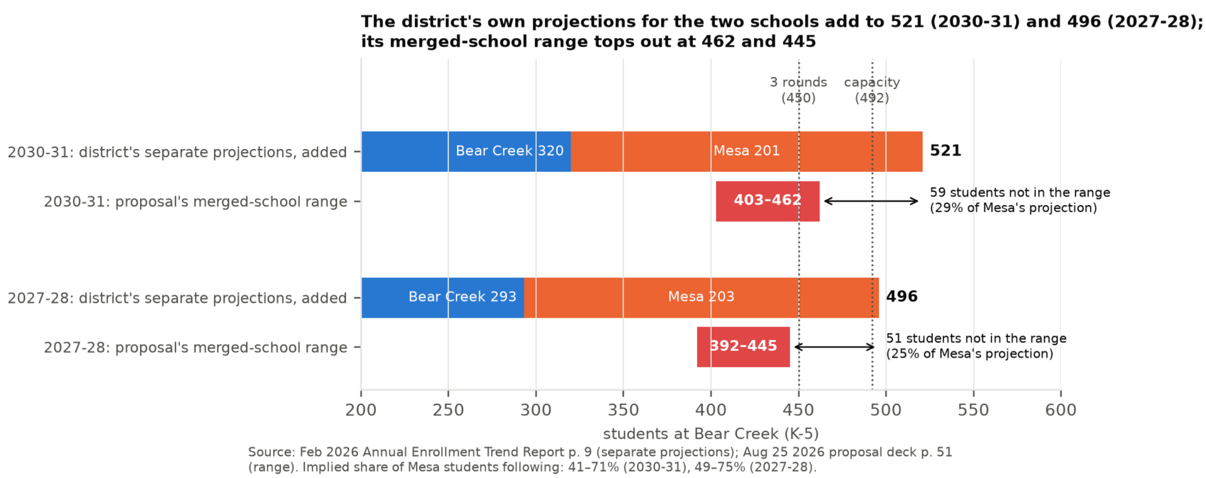
Summary

The district proposes an irreversible step on the basis of projections with no stated margin of error. Its own January 2026 projections for Bear Creek and Mesa add to 521 students in 2030–31; the Bear Creek building holds 492; the proposal's merged-school range tops out at 462. The gap is an assumption about how many Mesa families leave, or about how much open enrollment the district will cut, and the proposal does not say which. The district's projections for these two schools have moved 10–14% between successive runs, and closing Mesa cannot be undone while keeping it open can.



The district's own separate projections for the two schools (blue plus orange) exceed its merged-school range (red) by 59 students in 2030–31 and 51 in 2027–28. No model is involved. Feb. 2026 report p. 9; deck p. 51.

The numbers that matter (page in parentheses; n = number of school-year cases):

- Bear Creek's 2029–30 projection rose from 272 to 310 (+14%) and Mesa's fell from 224 to 202 (−10%) between the January 2025 and January 2026 runs; Bear Creek's 2028–29 projection went 248, 274, 288 in three consecutive runs, +16% over two years (Feb. 2024 p. 11; Feb. 2025 p. 9; Feb. 2026 p. 9).

- Across all BVSD elementary schools, one two-year-ahead projection in ten misses by more than 13% (n = 29); one year ahead, by more than 9% (n = 59). Four in five revisions to targets four to five years out are within about 12%, and the largest is 46% (n = 59 school-run pairs).

- An independent model built on the district's own method and grade counts reproduces the district's Bear Creek figure for 2030–31 (320; the model gives 317) only if kindergarten intake stops falling and holds at its 2023–25 level. Under that assumption the two schools together are about 523 and the merged school (90% of Mesa students following) has a central estimate of 503 with a 57% chance of exceeding 492. If intake keeps falling, the central estimate is 430 and the chance is 16%. The 73-student gap between the two assumptions is wider than the district's entire 59-student range.

- The proposal's range can be reached from the district's own numbers only by assuming either that 29–59% of Mesa's projected students go elsewhere, or that open enrollment into the merged school is cut to near zero. The documents do not say which.

- The deck's own resident-student figures for the combined area are 503 today, 473 in 2027–28 and 522 in 2030–31 (p. 51), while its chart of 2030 residents reads about 445 (p. 44). No document reconciles them.

- The deck gives package-level savings of $3.5–4.0M a year against one-time costs of $7.5–10.0M plus $5.0M, with no Mesa-specific figure (p. 60).

Conceded at the outset. Mesa is small; it has been on the district's enrollment-advisory list every year the list has existed; and the district's central projection for it is reasonable. The independent model here agrees that Mesa stays small under either assumption. The question is not whether Mesa is small but whether the merged school fits the building, and on that the documents presented do not carry the evidence an irreversible decision requires. The independent model's central estimate under one assumption (430) lies inside the district's range; that is not the finding. The finding is the chance of ending above 450 and above 492, and the fact that an unstated modeling choice moves the answer by more than the district's whole range.

Terms used throughout. *Run*: the district's annual projection (January, published in the February trend report). *October count*: the state-mandated October enrollment count, K–5. *Merged school*: Bear Creek after Mesa's students are added. *Round*: one class per grade, about 150 students K–5; three rounds about 450; Bear Creek's 492 seats are 3.5 rounds, or 21 classrooms. *Trend assumption*: kindergarten intake keeps falling on its 2014–25 trend. *Level assumption*: kindergarten intake holds at its 2023–25 average. *Share following*: the fraction of Mesa's projected students who enroll at Bear Creek. *Capture rate*: the share of an area's resident students who attend their neighborhood school. *Central estimate*: the middle of the simulated outcomes. *80% range*: the span that contains the outcome four times in five; its ends are the 1-in-10 low and 1-in-10 high.

The decision and the numbers behind it

On August 25, 2026 the district presented its Resilient Schools Proposal; the Board discusses it on September 8 and acts on September 22, with changes effective in 2027–28 (deck p. 3). For South Boulder it proposes to "close Mesa and merge the school communities in the Bear Creek building", to "redraw attendance boundaries to combine the Bear Creek, Mesa and dual attendance areas", and to move Mesa's RISE program to Bear Creek (p. 49). The stated rationale (p. 50) is that the consolidation "reduces excess capacity and boosts enrollment at the receiving school to 3-rounds"; that Mesa "is projected to continue to decline" and has "the second lowest number of resident students in its attendance area"; and that Bear Creek is at 63% utilization (enrollment as a share of capacity) and 2.1 classes per grade against Mesa's 54% and 1.5. Table 1 reproduces the deck's figures for the consolidation.

The proposal’s figures for the consolidation (deck p. 51, text layer).

	2025–26 (actual)	2027–28 (proj.)	2030–31 (proj.)
Bear Creek: capacity / residents / enrolled	492 / 275 / 312		
Mesa: capacity / residents / enrolled	418 / 228 / 224		
Merged school at Bear Creek: resident students		473	522
Merged school at Bear Creek: enrolled students		392–445	403–462
Merged-school utilization		80–90%	82–94%

The deck’s footnote reads: “Projected enrollment ranges dependent on number of resident students from the sending school attending the receiving school.” It states the dependence; it does not state the share, and no probability is attached to either end.

District-defined thresholds used throughout: one round is about 150 students, three rounds about 450 (Oct. 21, 2025 work session, slide 8); the Long Range Advisory Committee’s advisory status is ≤ 2 rounds and ≤ 60% of capacity, and its community-engagement status ≤ 1.5 rounds and ≤ 50% (Feb. 2026 report p. 10). Mesa has appeared on the advisory list in every year it has been published (Feb. 2024 p. 13; Feb. 2025 p. 11).

Two figures that circulate publicly differ from the documents. Mesa’s October 2025 count is 224 in every BVSD table (CDE’s membership count is 225). Bear Creek’s “program capacity” is 467 (3.0 rounds) in the January 2024 table and 492 (3.5 rounds) in every later table. The documents reviewed do not explain the change. On its own it moves Bear Creek’s stated utilization by five points.

The evidence standard

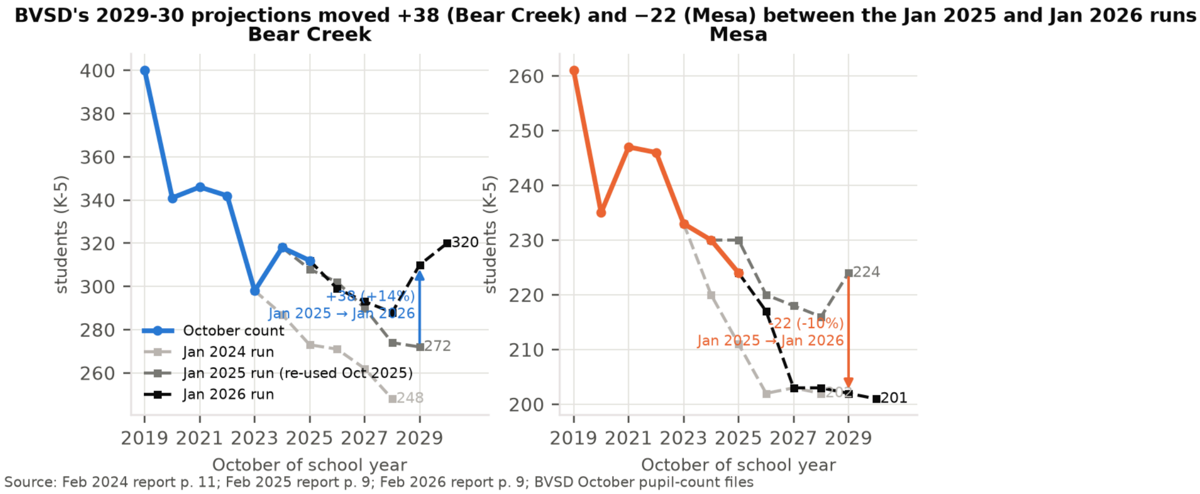
An irreversible decision should be held to a higher bar than a reversible one. Closing a school cannot be undone; keeping one open can. Before approving any closure the Board should have in hand:

1. a projection with stated prediction intervals, back-tested against the district’s own October counts;
2. the retention, resident-capture and open-enrollment assumptions behind any merged-school projection, stated explicitly;
3. a demonstration that the merged school fits the receiving building with margin under the upside enrollment cases, not only the central one. The fit should be shown net of the classrooms used by center-based special-education programs;
4. a school-specific accounting of one-time costs, recurring savings, and revenue at risk from families who leave the district;
5. a transition plan for RISE, school-age care, preschool and transportation.

Sections 3–8 work the enrollment evidence and show that (i)–(iii) are not met by the documents presented. Section 10 shows that (iv) is not met. Section 11 lists the open questions under (v). The standard applies to every component of the package; this appendix covers the one its author can speak to.

How much the district’s own numbers have moved

The district’s projections for these two schools have moved 10–14% between successive runs and 16% over two runs, and the latest run is the first to turn Bear Creek upward, for reasons the documents do not give. Figure 2 shows the three runs against the October counts; Table 2 gives the numbers.



BVSD’s projections for Bear Creek and Mesa by run, against the October counts. The January 2026 run is the first in which Bear Creek’s path turns upward. Feb. 2024 p. 11; Feb. 2025 p. 9; Feb. 2026 p. 9.

BVSD projections by run and target year (K–5). Feb. 2024 p. 11; Feb. 2025 p. 9 (re-printed as Oct. 2025 work session slide 17); Feb. 2026 p. 9; October counts from the pupil-count files.

School	Target year	Jan 2024 run	Jan 2025 run	Jan 2026 run	October count
Bear Creek	2024–25	287	318	–	318
	2025–26	273	308	312	312
	2026–27	271	302	299	
	2027–28	262	290	293	
	2028–29	248	274	288	
	2029–30	–	272	310	
	2030–31	–	–	320	
Mesa	2024–25	220	230	–	230
	2025–26	211	230	224	224
	2026–27	202	220	217	
	2027–28	203	218	203	
	2028–29	202	216	203	
	2029–30	–	224	202	
	2030–31	–	–	201	

1. The October 2025 work session, which introduced the Board to the South Boulder question, contained no new projections: its slide 17 is number-for-number the January 2025 table.

2. Between the January 2025 and January 2026 runs, Bear Creek’s 2029–30 projection rose by 38 students (+14%) and Mesa’s fell by 22 (−10%). Bear Creek’s 2028–29 projection rose in each of three successive runs, from 248 to 274 to 288.

3. The January 2026 path for Bear Creek is the first to turn upward (299, 293, 288, then 310, 320). The documents reviewed do not say why. The one boundary change adopted for 2026–27 converts the Bear

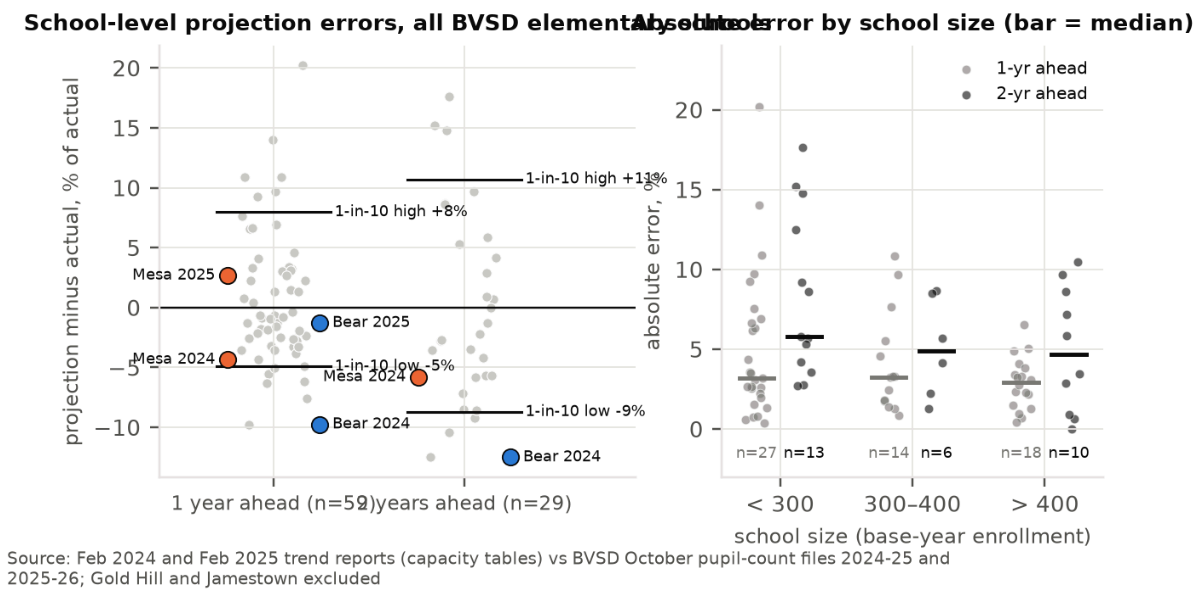
Creek/Creekside dual area to Bear Creek only. That area holds 37 elementary students, 23 of whom already attend Bear Creek (Sept. 9, 2025 boundary study p. 22), so it does not account for a 38-student revision. Section 5 suggests a likelier reason: kindergarten intake stopped falling.

4. The proposal’s merged-school range sits below the sum of the district’s own separate projections (Figure 1): 521 in 2030–31 and 496 in 2027–28 against ranges topping out at 462 and 445.

5. The deck’s resident counts for the combined area run 503 today (275 + 228, p. 51), 473 in 2027–28 and 522 in 2030–31, a 10% rise in three years in a district that attributes its decline to low births and an aging population (FAQ p. 6; deck p. 10). The deck’s own chart of 2030 projected residents (p. 44) shows markers at about 268 for Bear Creek and 177 for Mesa, about 445 in total. Whether the figures share one definition is not stated; no document reconciles them.

How accurate have the district’s school-level projections been?

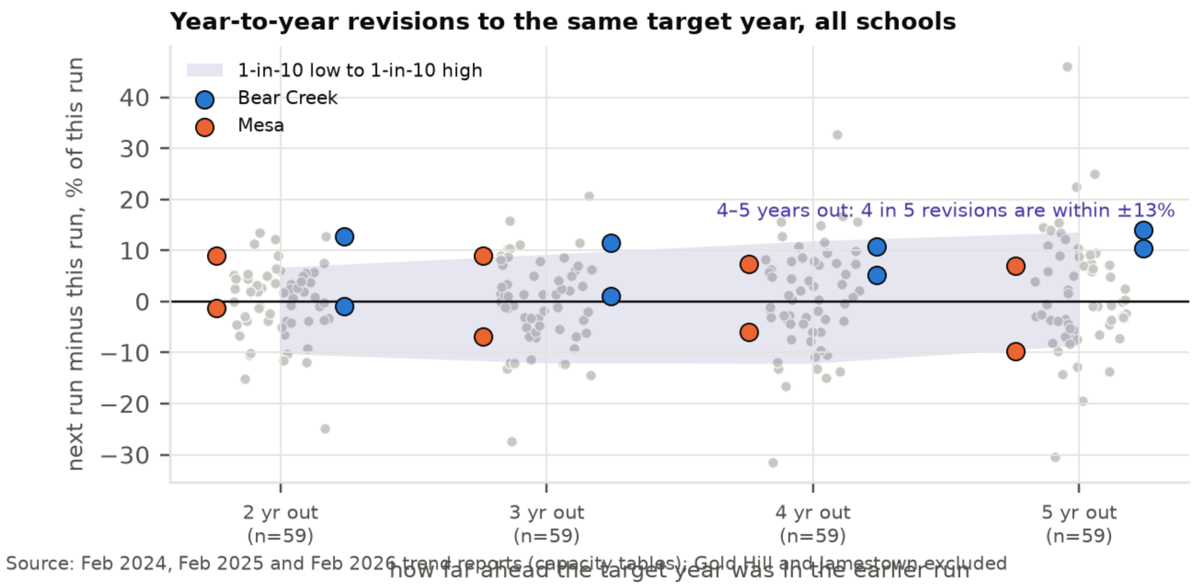
One year ahead, one school-level projection in ten misses by more than 9%; two years ahead, by more than 13%. The district has leaned neither high nor low. With three runs and official counts through October 2025, the projections can be scored one and two years ahead for every elementary school in the tables: 59 one-year cases and 29 two-year cases, excluding the two tiny mountain schools (Gold Hill and Jamestown; University Hill is absent from the January 2024 table). Figure 3 and Table 3 summarize the errors; Figure 4 shows revisions between successive runs to the same target year, the only available evidence on uncertainty three to five years out.



Left: percent error of BVSD’s school-level projections by horizon, all elementary schools, with the 1-in-10 low and high marked; Mesa and Bear Creek highlighted. Right: absolute error by school size.

Absolute percent error of BVSD school-level projections, and magnitude of revisions to the same target year between successive runs. The two-year cases all target October 2025; the revision rows are the same 59 school-run pairs seen at four horizons.

	n	Median	80th pct.	90th pct.	Max	Average error (sign kept)
Error, 1 year ahead	59	3.2%	6.4%	9.4%	20.2%	+0.6%
Error, 2 years ahead	29	5.7%	9.4%	13.0%	17.6%	0.0%
Revision, target 2 years out	59	3.9%	9.1%	11.5%	24.9%	
Revision, target 3 years out	59	6.0%	10.8%	12.3%	27.4%	
Revision, target 4 years out	59	6.2%	12.2%	15.1%	32.8%	
Revision, target 5 years out	59	6.5%	12.5%	14.7%	46.1%	



Revisions between successive runs to the same target year, by how far out the target was in the earlier run. Four in five revisions to four- and five-year targets are within about 12%.

Three points bear on the decision. First, district-wide totals are projected well (errors of −1.1%, −0.7% and +1.4% for the three total-level cases; the district's own December 2025 update reports the 2025–26 count came in "1% more than projected", news article p. 1). The school-level errors are not a sign of a poor model; they are what school-level projections look like when resident moves, open enrollment and small numbers all contribute. Second, the largest two-year misses were at schools under 300 students (80th percentile 14% against 9% for larger schools), and Mesa is among the smallest. Third, the average error is near zero: the district has been about as likely to be too low as too high, which is the situation in which an irreversible choice is costly.

For the two schools at issue, the January 2024 run projected Bear Creek at 287 for October 2024 and 273 for October 2025; the actuals were 318 and 312 (−9.7% and −12.5%). Mesa was projected at 220 and 211 against actuals of 230 and 224. The January 2025 run did well one year out (Bear Creek −1.3%, Mesa +2.7%). The point is not that the district projects badly; it is that a five-year point projection for a 220-student school carries a band of several tens of students, and the proposal presents none.

Two cautions. The 29 two-year cases all score one run (January 2024) against one October (2025), a year the district itself reported as "steeper than projected" (Feb. 2026 p. 5), so they are not 29 independent tests. And using revisions as a guide to uncertainty at longer horizons assumes that the district's later revisions are not systematically reversed by later ones; on that assumption the revision magnitudes are a lower bound on the error at those horizons.

An independent projection with a stated range

An independent model that follows the district's own method but reports a range puts the merged school anywhere from about 360 to about 580 in 2030–31, depending mainly on one assumption the district has not stated.

Method in brief

The model follows the district's stated method (cohort survival, Feb. 2026 p. 7): each grade's October count is carried forward to the next grade at the rate observed historically, and a new kindergarten class is added each year. Twelve years of K–5 counts for each school give eleven grade-to-grade rates for each of grades 1–5 and twelve kindergarten intakes. Future years are simulated 10,000 times. In each simulated year, one historical year is drawn at random and its kindergarten surprise and all five grade-to-grade rates are applied to both schools at once, so a shock like 2020 stays one event and the two schools rise and fall together, as they do in the data. Fitted on data through 2023 and 2024 and scored on the counts that followed, the model's 80% range caught 72–76% of actual outcomes across 26 schools, as close to 80% as that many cases can show; its errors are the same size as the district's on the same school-years. It is not a better forecaster than the district and is not offered as one. Detail and the full backtest are in Appendix B.

The kindergarten assumption decides the answer

Kindergarten intake is the one place where a modeling choice is unavoidable, and it is made two ways, presented as co-equal throughout:

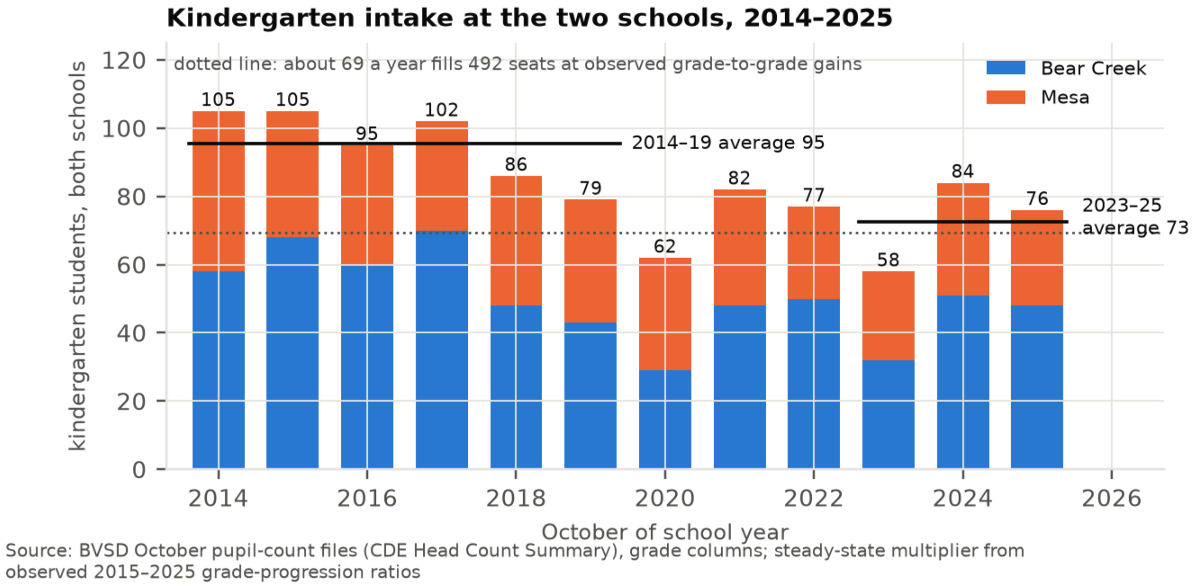
Trend assumption.

> Intake follows a straight-line trend (in logs) fitted to 2014–2025, with the uncertainty in that trend line included. This extrapolates the twelve-year decline.

Level assumption.

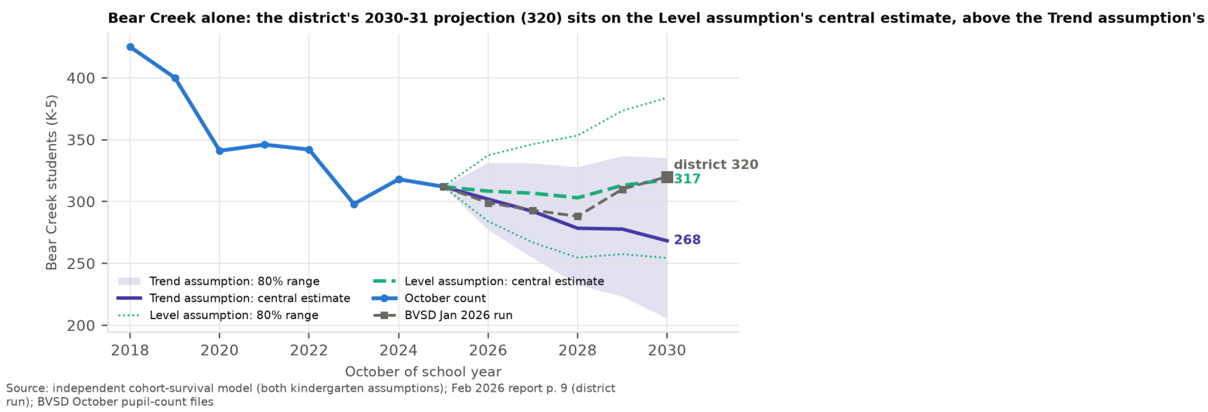
> Intake holds at its 2023–25 average, with the uncertainty in that average included. This assumes the decline has ended.

The evidence favors giving the level assumption at least equal weight. Combined kindergarten intake at the two schools was 105, 105, 95, 102, 86 and 79 in 2014–2019, stepped down to 62 in 2020, and has been 82, 77, 58, 84 and 76 since (Figure 5). A trend fitted through a step-then-plateau extrapolates a decline that, on this evidence, ended five years ago: under the trend assumption, Bear Creek's first simulated kindergarten class (central estimate 39) sits below four of the last five observed intakes (48, 50, 32, 51, 48). The district's December 2025 update reports that "the number of births in Boulder seems to be leveling after a steep decline until 2019" and that kindergarten "came in higher than projected" at 93% of births five years earlier for the second year running (news article p. 1); the February 2026 report assumes the births-to-kindergarten ratio is "relatively stable" (p. 8). In backtest the two assumptions err in opposite directions: the trend assumption under-projected and the level assumption over-projected (Appendix B), which is the reason to show both.



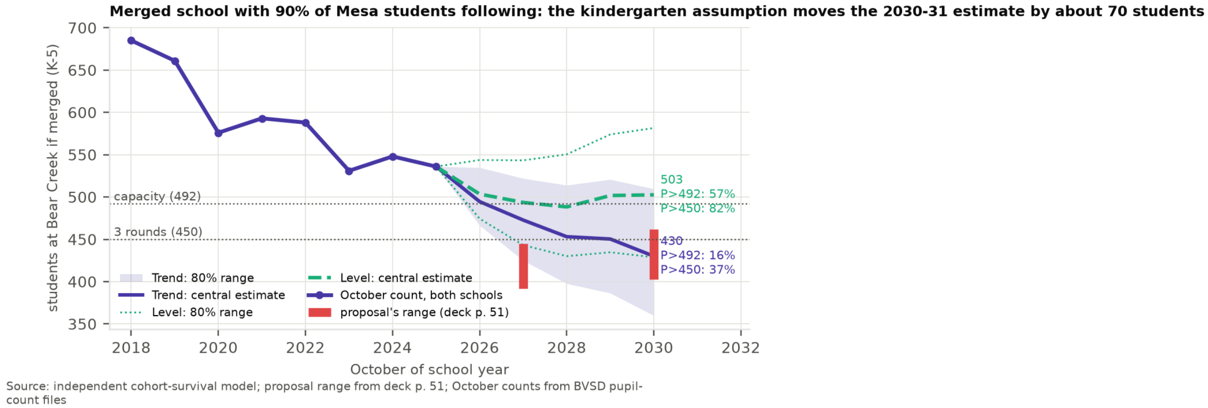
Kindergarten intake at the two schools fell from about 100 a year to about 75, then flattened after 2020. At the observed grade-to-grade gains, about 69 a year fills 492 seats in steady state; the 2023–25 average is 73. October pupil-count files.

There is a direct way to see which assumption the district's own run resembles. Its Bear Creek projection for 2030–31 is 320 (Feb. 2026 p. 9). The level assumption's central estimate for Bear Creek is 317; the trend assumption's is 268 (Figure 6). The district's path sits on the level assumption. Under that assumption the two schools together have a central estimate of 523 (the district's own sum is 521), above the building's 492 seats, and the published range of 403–462 is reachable only if 59–118 of those students go elsewhere.



Bear Creek alone under the two kindergarten assumptions, with 80% ranges, and the district's January 2026 run. The district's 320 for 2030–31 sits on the level assumption's central estimate (317), not the trend assumption's (268).

What matters for the Board is the size of the gap. With 90% of Mesa's students following, the merged school's 2030–31 central estimate is 430 under the trend assumption and 503 under the level assumption: 73 students, about three classrooms, more than the 59-student width of the district's entire range (Figure 7, Table 5). The level estimate is itself sensitive to the window: 505–547 for two- to six-year windows, and 484 if the year-to-year noise is centered on its median rather than its mean (Appendix B). Under every one of those variants the central estimate is at or above the top of the district's range. The documents do not say which assumption the district's own projection makes. Whether the merged school fits the building turns on it.



The merged school with 90% of Mesa's students following, under both assumptions, with 80% ranges, the proposal's range (red) and the 450 and 492 lines. The chance of exceeding 492 in 2030–31 is 16% under the trend assumption and 57% under the level assumption.

Results

Independent projection for 2030–31 (students, K–5). Low and high are the 1-in-10 low and 1-in-10 high. The check model is a random walk with drift on total enrollment (Appendix B).

	Low (1 in 10)	Central	High (1 in 10)
Bear Creek, trend assumption	206	268	335
Bear Creek, level assumption	254	317	384
Mesa, trend assumption	157	180	207
Mesa, level assumption	183	205	232
Both schools, trend assumption	377	448	529
Both schools, level assumption	448	523	604
Both schools, check model	377	461	566
BVSD Jan 2026 run, both schools		521	
Proposal's range, merged school	403		462

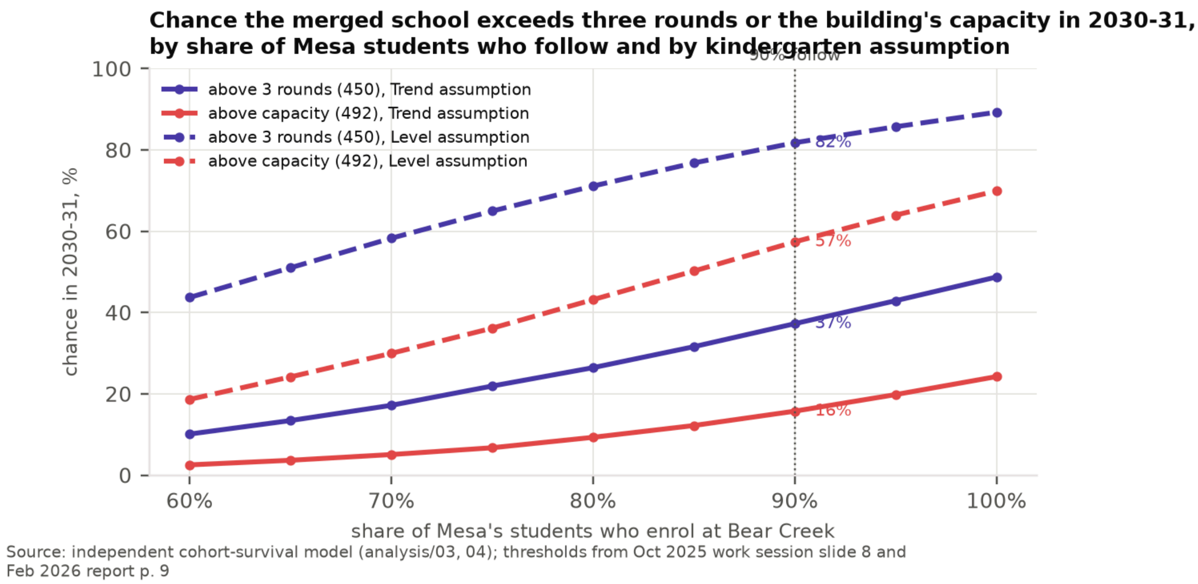
Merged school in 2030–31 by kindergarten assumption, 90% of Mesa's students following.

Assumption	Central	Low (1 in 10)	High (1 in 10)	Chance above 450	Chance above 492	Chance below 300
Trend	430	360	510	37%	16%	1%
Level	503	429	582	82%	57%	0%

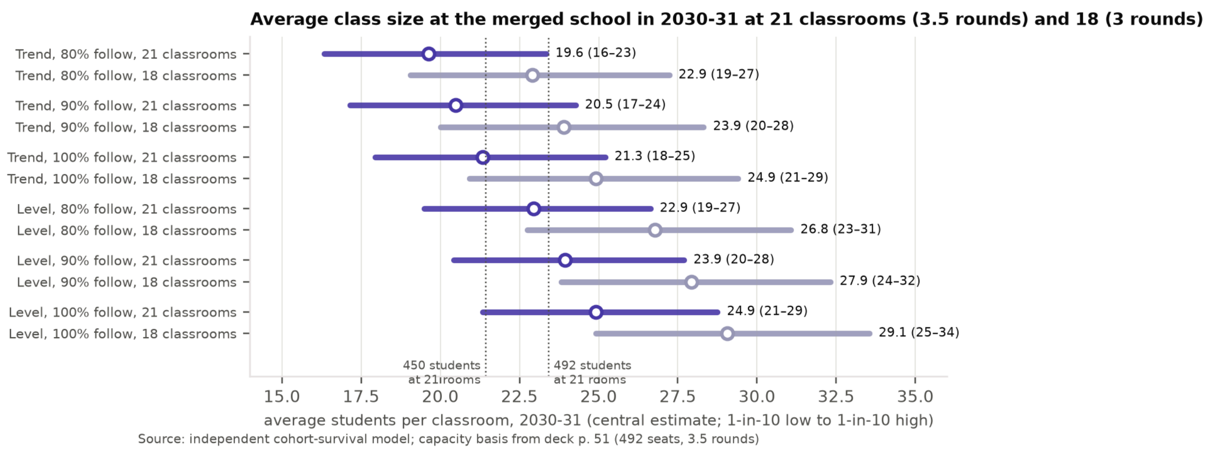
Under either assumption Mesa alone stays between about 160 and 230 students, below two rounds, and Bear Creek alone stays below three rounds. The model therefore agrees with the district that Mesa is small and will stay small. That is conceded. The dispute is about the merged school.

What the merged-school range assumes

Under either kindergarten assumption the risk is one-sided: the chance of a merged school below two rounds is at most 2%, while the chance of exceeding the building runs from 16% to 57% at 90% following. Let r be the share of Mesa's projected students who enroll at Bear Creek rather than elsewhere. Merged enrollment is Bear Creek's simulated path plus *r* times Mesa's, drawn jointly. Figure 8 gives the probabilities that matter across *r*; Figure 9 gives the implied class sizes.



Chance that the merged school exceeds three rounds (450) or the building's capacity (492) in 2030–31, by share of Mesa's students who follow, under both kindergarten assumptions. At 90% following: 37% and 16% (trend); 82% and 57% (level).



Average class size at the merged school in 2030–31 at 21 classrooms (the 3.5-round basis of 492) and at 18 (a three-round school). At 21 classrooms and 90% following, the central estimate is 20.5 (trend) or 23.9 (level); at 18, 23.9 or 27.9.

Under the trend assumption with 90% following, the 2030–31 central estimate is 430 (80% range 360–510), the chance of exceeding 450 is 37%, of exceeding 492 is 16%, and of falling below 300 is 1%. Under the level assumption the central estimate is 503 (429–582) and the chances are 82%, 57% and 0%. In the first year of the merged school, 2027–28, the central estimates are 473 and 494 and the chances of exceeding 492 are 30% and 52%. The chance-below-300 column shows that the downside of a too-small merged school is negligible under either assumption; the risk is toward overcrowding.

Two readings of 403–462, and both are a problem

The district's separate January 2026 projections are 320 and 201 for 2030–31 and 293 and 203 for 2027–28. Read on those, the ranges 403–462 and 392–445 imply that r lies between 0.41 and 0.71 in 2030–31 and between 0.49 and 0.75 in 2027–28: the top of the range assumes that between a quarter and three-fifths of Mesa's projected students go elsewhere.

The district may reply that the range is defined on residents, not on the two schools' enrolled counts, and that standalone projections are not additive because each contains open-enrollment inflow. Read that way, the range is a fork. The district projects 522 residents in the combined area in 2030–31 (p. 51). If Bear Creek's current 95 open-enrolled-in and out-of-district students stay, 403–462 requires a resident capture rate of 59–70%, below Mesa's 68% and Bear Creek's 79% today (p. 44 bar labels; p. 51); if Mesa's own 68 open-enrolled students are also counted, 46–57%. If instead capture stays near today's rates, the range requires open enrollment into the merged school to be cut to near zero. The FAQ says the latter is contemplated where "new short term enrollment will push the school near capacity": projections then "take into account managing choice enrollment beginning in 2027–28 to ensure that resident students can be accommodated" (p. 3). It does not say whether Bear Creek is one of those cases.

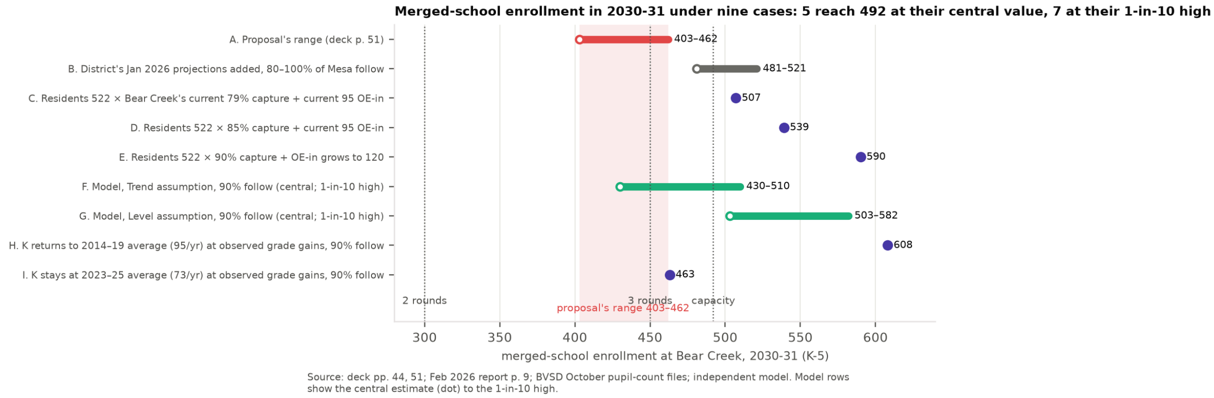
Either branch is a problem for the proposal. One assumes South Boulder families leave their neighborhood school at rates below anything observed today. The other assumes the district will cap the very choice enrollment that its own rationale says a fuller, three-round school should attract (pp. 50, 60–61). And the second branch has a corollary the Board should weigh: managing choice is the district's only lever against overcrowding at Bear Creek, and pulling it moves the merged school back toward two and a half rounds. The risk is two-sided, and the lever moves the school between the sides. The right question for staff is not "which reading is correct" but "what share following, what capture rate and what open-enrollment level does 403–462 assume, and does it assume choice management at Bear Creek".

RISE, AIM and effective capacity

Bear Creek's 492 seats are "program capacity based on current use of the building" (Feb. 2026 p. 9, footnote), so its existing programs are inside that figure. The bvsd.org proposal page states that "all existing AIM and RISE programs would be housed at Bear Creek" and the deck that "Mesa's RISE Program moves to the Bear Creek building" (page p. 7 of the saved copy; deck p. 49); the FAQ says program space "was considered" (p. 3). None of the documents states how many classrooms these center-based programs occupy at either school or whether 492 nets out the incoming one. Table 7 (Appendix C) recomputes the exceedance chances at effective capacities of 492, 470 and 445, roughly one and two classrooms removed; it is illustrative pending the district's statement. At an effective capacity of 445, the chance of exceeding it in the first year with 90% following is 79% (trend) or 89% (level).

What would it take for Bear Creek to fill? Upside cases

The district attributes the decline to low births, an aging population and constrained housing turnover (FAQ p. 6; deck p. 10). Each driver can also reverse, and the merged school starts close to the thresholds. Figure 10 lays out nine cases for 2030–31: five reach 492 at their central value, seven at their 1-in-10 high.



Merged-school enrollment in 2030–31 under the proposal's range (A), the district's own projections added (B), three resident-capture cases (C–E), the two model assumptions (F–G), and two steady-state kindergarten cases at observed grade-to-grade gains (H–I).

Leveling-off (G, I).

> At the observed grade-to-grade gains (each kindergarten class grows about 31–35% by fifth grade), a steady intake of about 69 a year fills 492 seats; the 2023–25 average is 73, which at 90% following gives 463 (I). The level assumption is this case with noise: central estimate 503, 1-in-10 high 582 (G).

Resident capture (C–E).

> At Bear Creek's current 79% capture rate plus its current 95 open-enrolled-in and out-of-district students, the merged school would have about 507 in 2030–31; at 85% capture, about 539 (Section 6.1 shows why the proposal's range requires capture to fall to 59–70% or open enrollment to be cut).

Open enrollment.

> The proposal's stated aim is that a three-round school with fuller programs is a better student experience (deck pp. 60–61). If so, it will attract open enrollees, and every 25 is one classroom. Bear Creek currently has 82 in-district open-enrolled students and 13 out-of-district students; the district does not publish which schools they come from.

The district's own numbers, unadjusted (B).

> If the January 2026 projections hold and all of Mesa's students follow, the merged school has 521 students, 29 above capacity.

Housing turnover.

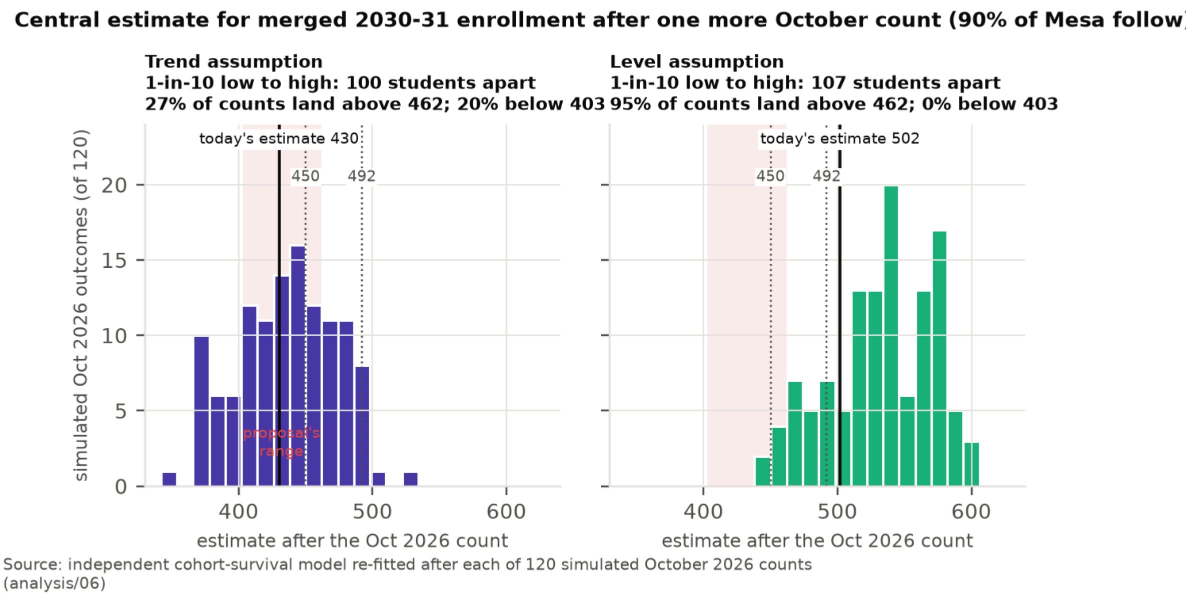
> The district's boundary study uses a yield of 58 elementary students per 324 single-family dwellings, 0.18 per home (Sept. 9, 2025, p. 22). Moving the trend-assumption central estimate of 430 to 492 requires about 62 students: about 350 homes turning over at that average yield, or 62 homes passing to households with one elementary-age child each. The FAQ describes the over-60 population as the fastest-growing segment and "aging in place" as a constraint on turnover (p. 6; deck p. 10). No age or housing data for the two areas is in the record, so this case is illustrative arithmetic, not a forecast.

None of these cases is a prediction. They show that outcomes at or above the building's capacity require nothing exotic: a continuation of the last three years' kindergarten intake, or the combined area behaving the way Bear

Creek's area already does. The district's reply that over-enrollment has reversible remedies (choice limits, boundaries, portables) is correct, and it is the point: every one of those remedies costs money the deck's one-time line does not count, and Meadowlark already relies on four portables adding about 100 seats (Feb. 2026 p. 9, footnote).

What the next October count would change

The October 2026 count will be the first taken under the 2026–27 boundaries, and the January 2027 run the first that could reflect them and the proposal's open-enrollment rules. No projection can be back-tested against a portfolio that has not yet existed, so standard (i) cannot be met for the merged school without that count. The count also matters in size. Re-running the model after each of 120 simulated October 2026 counts, the central estimate for the merged school in 2030–31 moves across a range of roughly 100 students from the 1-in-10 low to the 1-in-10 high under either assumption (Figure 11): about two classrooms in either direction. Under the trend assumption, 27% of simulated counts push the estimate above the top of the district's range and 20% below its bottom. This sensitivity is largely the sensitivity of the trend estimate to a single kindergarten cohort; it is a statement about how little one more year of data is needed to move the answer, not a claim that waiting resolves the question. The expected width of the 80% range narrows only modestly (from about 150 students to about 128 under the trend assumption, and hardly at all under the level assumption).



Where the central estimate for the merged school in 2030–31 would land after each of 120 simulated October 2026 counts, under both assumptions, against the proposal's range (shaded) and the 450 and 492 lines.

The cost of meeting the standard is one enrollment cycle: the Board's September 22 action is tied to the open-enrollment window that opens November 1 (deck p. 67), so a decision made after the October 2026 count would take effect in 2028–29 at the earliest.

What no projection captures

Every projection in this appendix, the district's and ours, is fitted on grade-progression rates and open-enrollment flows observed under the current Boulder portfolio and choice rules. The proposal changes both in 2027–28. It closes Douglass, Flatirons and Birch and the Community Montessori building. It relocates High Peaks and Community Montessori. It redraws attendance areas across the region (deck pp. 46, 52, 23, 59). And it gives students of closed or moved schools a one-year open-enrollment priority (deck p. 63; FAQ p. 2, “The Board will study this item further on Sept. 8”). Rates fitted under the old portfolio do not carry across a change of that size.

The deck assigns no closed school’s residents to Bear Creek other than Mesa’s: Douglass’s area goes to Coal Creek, Eisenhower and Heatherwood (p. 46), Flatirons’ to Foothill and Whittier (p. 52), Birch’s to Kohl (p. 25). After the package, the Boulder-region neighborhood schools the deck projects at or near three rounds in 2030–31 are Foothill (484–492, p. 54) and Bear Creek at the top of its range (403–462, p. 51); district-wide, the deck reports the number of schools below two classes per grade falling from 14 to 6 (p. 58). Among options studied and not proposed was closing Creekside and redistributing its students “to Mesa, Bear Creek and Eisenhower” (p. 56). The FAQ’s statement that choice enrollment will be managed at schools nearing capacity (p. 3, quoted in Section 6.1) is the only text on re-sorted flows; no document states whether 403–462 includes them or names Bear Creek’s sending schools.

The identifiable channels point toward more students at Bear Creek: it becomes one of very few Boulder-region schools at or near three rounds, which the district describes as the design (p. 43: 16 schools where 10 would sustain three classes per grade); the proposal’s own claim is that fuller programs attract enrollment; and choice priority is being extended to displaced families across the region. The direction is not certain and the district has not modeled it. No flow assumptions are invented here.

Other sources of variance that no projection carries, each with its documentary support or labeled unquantified:

- **Timing of the choice window.** Open enrollment for 2027–28 runs Nov. 1 to Dec. 18, 2026, after the vote (deck p. 67). Mesa families who leave in that window lower the 2026–27 base to which any retention share is applied. Unquantified.

- **Future use of vacated buildings.** “Decisions about the future use of buildings are not part of the Resilient Schools proposal” (FAQ p. 3). Whether another operator occupies the Mesa building is outside the record; unquantified.

- **Housing turnover in an aging stock.** FAQ p. 6; Section 7. Unquantified.

- **Preschool placement and kindergarten capture.** BVSD’s K–5 tables exclude preschool by design (deck p. 12 footnote); CDE separately records 13, 20 and 25 pre-kindergarten students at Mesa in 2023–24 to 2025–26. The proposal says preschool “will continue to be offered in the area” without a location (deck p. 64). The effect of preschool placement on kindergarten capture is addressed in no document. Unquantified.

Cost and benefit

The deck gives package-level savings and costs only; there is no Mesa-specific figure on either side, and the range’s own leakage assumption puts revenue at risk. The district’s figures (deck p. 60, for the whole package) are recurring annual savings of $3.5–4.0M, “estimated funding redirected to support the student experience every year”, against one-time transition costs of $7.5–10.0M for facility modifications (“some offset by savings from Bond projects”) and $5.0M for school transition support (“budgeted, FY27”). No per-school breakdown, no figure for maintaining vacant buildings, no portable cost, no RISE or AIM relocation cost and no transportation cost appears in the deck, the FAQ or the bvsd.org pages. Two one-time figures circulate publicly: “up to $10M” is the facility line alone and “$12.5–15M” is the sum of both lines; both are consistent with p. 60.

Arithmetic on the district’s figures: simple payback of the one-time cost from the recurring savings is 3.1 to 4.3 years (12.5–15.0 divided by 3.5–4.0), if the savings are realized in full; the district would note that the $5.0M is already budgeted and part of the facility line is offset by bond funds. There is no Mesa-specific line on either side.

The benefit is a projection that may not be realized in full; the one-time cost is undercounted if the merged school needs portables or further boundary work. On the revenue side there is evidence from elsewhere, with a caveat. A study of all California public school districts, 2011–2019 (Pearman, Stanford, May 2026; `data/raw/literature/`), estimates that districts entering a closure regime reduced per-pupil expenditures by about $447 and per-pupil revenues by about $433. Both estimates are imprecise and not statistically significant, but they are nearly equal, so districts' funding deficits and probability of a balanced budget were unchanged (abstract; p. 12). Closure regimes lost 287 students on average with "no commensurate effects on the number of teachers, principals, or other district staff" (abstract). The caveat is that Pearman's channel is students leaving the district under California's funding formula; BVSD's range concerns Mesa residents choosing among BVSD schools with first priority (deck p. 63), most of whom stay in the district's count. The relevance is narrower: the district's own range assumes that 29–59% of Mesa's projected students do not follow to Bear Creek, and the deck does not say where they go or what funding, transportation or program cost follows them.

Open questions the documents do not answer

1. How many classrooms do AIM and RISE occupy at Bear Creek today, and how many after Mesa's RISE program moves? Does 492 net them out? (deck p. 49; bvsd.org page p. 7)

2. What share of Mesa's students following, what resident capture rate and what open-enrollment level does 403–462 assume, and does it assume choice management at Bear Creek? (deck p. 51; FAQ p. 3)

3. Where do the 29–59% of Mesa's projected students that the range assumes do not follow go, and what funding and cost follow them? (deck p. 51; Feb. 2026 p. 9)

4. What reconciles the combined area's resident counts of 503 today, 473 in 2027–28 and 522 in 2030–31 (p. 51) with the chart showing about 445 residents in 2030 (p. 44)?

5. School-age care "will be offered at Bear Creek" (bvsd.org page p. 7); with what capacity relative to two schools' current demand? (FAQ p. 5 notes expansion is limited by staffing ratios and space.)

6. Preschool "will continue to be offered in the area"; where? Locations "will be shared by Oct. 1", after the vote (deck p. 64; FAQ p. 4).

7. The K–2 / 3–5 reconfiguration of Mesa and Bear Creek is listed among "other options studied" (deck p. 56); in October its fiscal impact was described as "uncertain" (work session slide 45). No comparison with the adopted option has been published.

8. Transportation: the deck states the eligibility rule (1.5 miles or more from the assigned school, p. 64) but not how many Mesa-area addresses fall outside that radius from Bear Creek.

Limitations and conclusion

Three projection runs exist, so direct accuracy is measured only one and two years ahead; the three-to-five-year figures are revisions, a lower bound on uncertainty under a stated assumption. The independent model uses only BVSD's grade counts; it has no births, housing or open-enrollment inputs, its 50% range is too narrow in backtest, and its two-year upper tail is thin, which if anything understates the chance of exceeding capacity. The share following and the open-enrollment response are assumptions shown as ranges. The effective-capacity table is illustrative. The turnover case uses a district yield figure from a different neighborhood. Costs of either error are not quantified because the district's documents do not quantify them. Capacity is the district's "program capacity based on current use of the building" and is itself a policy variable.

The documents presented do not contain a projection with a margin of error. They do not state the assumptions about following, capture or open enrollment behind the merged-school range. They do not show that the merged school fits the building under the higher cases once program space is counted. They contain no Mesa-specific cost or revenue figure. And they leave the transition questions in Section 11 open. The Board should decline to approve the Mesa/Bear Creek component of the Resilient Schools proposal at this time, and direct staff to return with an analysis that meets this standard.

Reproducibility

All scripts run from the repository root in numbered order: `scripts/` (extraction and verification) and `analysis/01_descriptive.py` through `06_value_of_waiting.py`. Figures are written to `figures/`, tables to `analysis/output/`. Random seeds are fixed. The source of this document is `report/report.tex`; `report/build.sh` renders it.

Data, verification, model detail and backtest

Data.

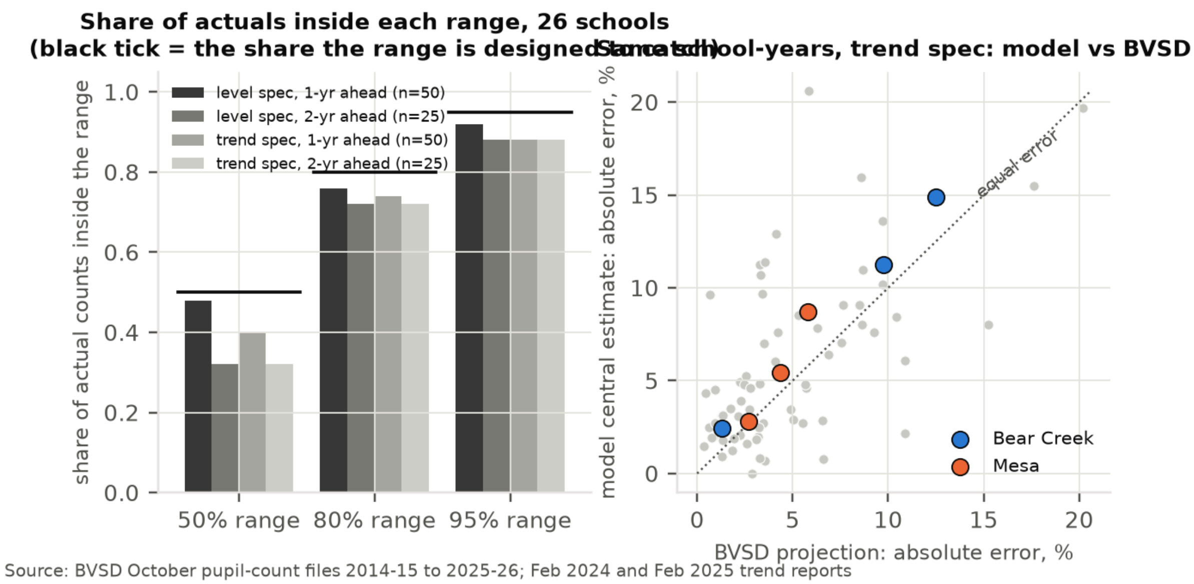
Every number traces to a page of a BVSD or Colorado Department of Education document held unmodified under `data/raw/`: the Aug. 25, 2026 deck (75 pp.); the February 2024, 2025 and 2026 Annual Enrollment Trend Reports; the Oct. 21, 2025 work session; the March 11 and Sept. 9, 2025 boundary items; BVSD's October pupil-count files for 2014–15 through 2025–26; CDE pupil-membership files for 2019–20 through 2025–26; the bvsd.org Resilient Schools page and FAQ; and the district's Dec. 12, 2025 enrollment news article. The capacity tables in the trend reports and work session are images; they were OCR'd, every Mesa and Bear Creek cell was checked by eye against the rendered page, and every other school's row was accepted only if the table's own current-year enrollment matched the official October count. Grade counts sum exactly to published head counts for every school in all 12 years. CDE's K–5 membership agrees with BVSD's funded count to within one or two students in every year. Full detail: `data/VERIFICATION.md`; disagreements between sources: `data/CONFLICTS.md`.

Model detail.

Grade-progression rates $g_{i,t} = N_{i,t}/N_{i-1,t-1}$ for grades 1–5 and years 2015–2025. Trend assumption: $\log K_t = a + bt + e_t$ fitted by least squares on 2014–2025, with (a, b) resampled by a pairs bootstrap for each simulated path and e drawn from the fitted residuals. Level assumption: $\log K_t = \log \bar{K} + e_t$, where $\bar{K}$ is the mean of a bootstrap resample of the 2023–25 intakes (one resample per path) and e is the fitted residual series centered on its mean, so that the expected intake equals the window mean. In each simulated year one historical year is drawn and its residual and all five rates are applied to both schools. Sensitivity of the level assumption for the merged school in 2030–31 at 90% following: central estimate 547 (two-year window), 503 (three), 519 (five), 505 (six); 484 with median-centered residuals (chance above 492: 45%). The check model is a random walk with drift on log total enrollment, drift and volatility bootstrapped from 2014–2025 log-differences, drift drawn once per path, with shared shocks across the two schools.

Backtest.

Fitted on data through October 2023 and October 2024, scored on the counts that followed, for the 26 elementary schools with complete 2014–2025 records (BCSIS and High Peaks summed to match the district's table). Coverage of the nominal 50 / 80 / 95% ranges: trend assumption 40 / 74 / 88% one year ahead (n = 50) and 32 / 72 / 88% two years ahead (n = 25); level assumption 48 / 76 / 92% and 32 / 72 / 88%. Actuals above the model's 1-in-10 high at two years: 6 of 25 (trend), 4 of 25 (level); below the 1-in-10 low: 1 and 3. Median absolute error of the central estimate, one and two years: trend 3.3% and 8.4%, level 4.0% and 6.5%, against the district's 3.3% and 5.7% on the same school-years (n = 49 and 24 paired). Average signed error at two years: trend −0.6% (under-projects), level + 3.0% (over-projects). For Bear Creek, the trend assumption fitted through 2023 gave central estimates of 282 and 265 against actuals of 318 and 312, outside its 80% range both years; the level assumption gave 292 and 288, inside. The 80% coverage figures are not statistically distinguishable from 80% at these sample sizes, but the cases are not independent (all two-year cases target one October), so the backtest can fail to refute calibration rather than establish it.



Backtest across 26 schools: share of actual counts inside each range under both assumptions (left; black tick = designed coverage), and the model's errors against the district's on the same school-years (right).

Scenario and effective-capacity tables

Merged-school outcomes in 2030–31 by share of Mesa's students following, both assumptions.

Assumption	Following	Central	Low (1 in 10)	High (1 in 10)	Above 450	Above 492	Below 300
Trend	80%	412	343	490	26%	9%	2%
Trend	90%	430	360	510	37%	16%	1%
Trend	100%	448	377	529	49%	24%	0%
Level	80%	482	409	559	71%	43%	0%
Level	90%	503	429	582	82%	57%	0%
Level	100%	523	448	604	89%	70%	0%

Chance that merged enrollment exceeds an effective general-education capacity (illustrative; 90% and 100% of Mesa's students following).

Assumption	Year	Following	Central	Above 492	Above 470	Above 445
Trend	2027–28	90%	473	30%	53%	79%
Trend	2027–28	100%	493	51%	75%	89%
Trend	2030–31	90%	430	16%	26%	40%
Trend	2030–31	100%	448	24%	36%	52%
Level	2027–28	90%	494	52%	75%	89%
Level	2027–28	100%	514	73%	87%	96%
Level	2030–31	90%	503	57%	71%	84%
Level	2030–31	100%	523	70%	82%	91%